## COASTAL AREA STABILIZATION WITH MANGROVE FOREST

Though Kenya Forest Service (KFS), in collaboration with the Eden Reforestation Project, has planted over 3.8 million mangrove propagules, covering approximately 475 hectares in areas including Kipini, Ozi, Kilelengwani, and Chara within the Tana Delta Sub-County, several studies have documented the changes in mangrove cover in the Kipini area, highlighting both declines and subsequent recoveries over recent decades. Because of sediment shifts and weakened coastal protection the river delta is prone to erosion.

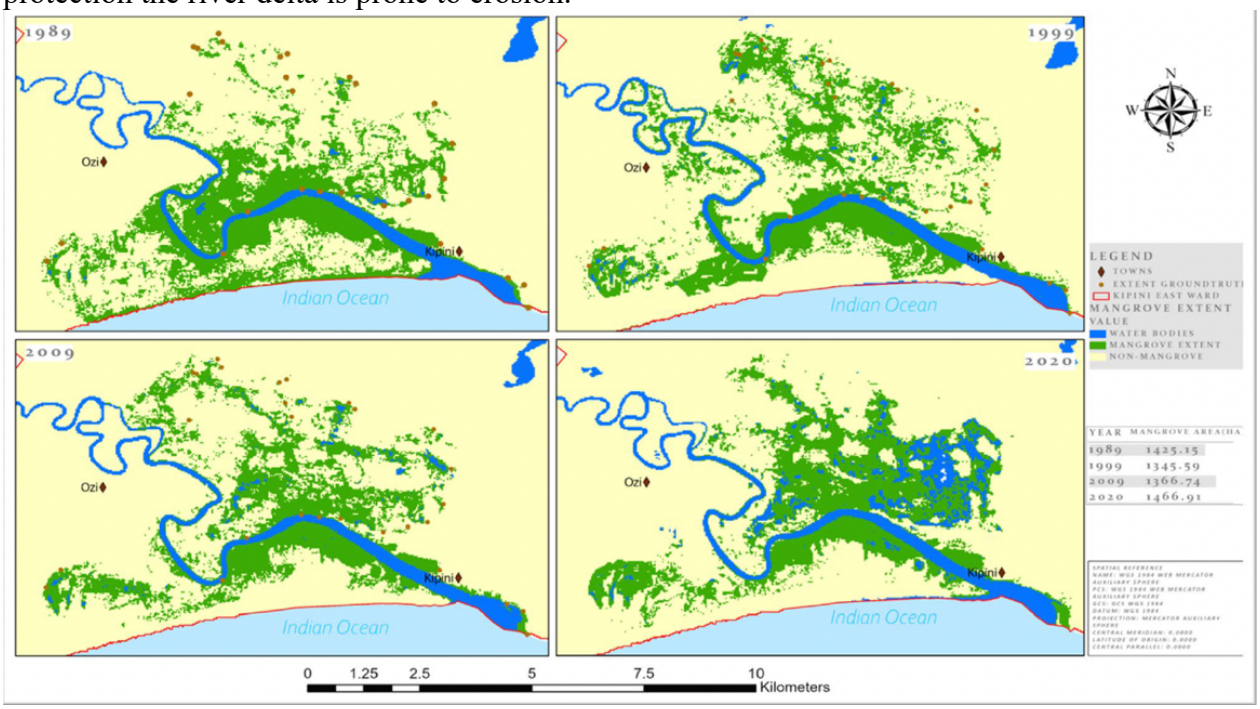



Figure: Temporal Changes in Mangrove Coverage in the Kipini Region (1989–2020) (springer.com)

**Site Assessment:**

As we can see from the geography of Kipini, this area is influenced by both fluvial inputs from the Tana River and tidal actions from the Indian Ocean. Kipini experiences semi-diurnal tides which can vary from 0.56 to 3.36 meters (timetides.org), with notable variations between low and high tides. The region has a mix of brackish water due to river outflow and saline ocean water, which impacts the growth of certain mangrove species, with salinity levels typically between 10 to 35 Practical Salinity Units (PSU) (mdpi.com). The coastal and deltaic zones around Kipini generally contain alluvial soil and sandy loams, supporting vegetation like mangroves (fao.org). Mangrove growth is impacted by rainfall variability, particularly during the long rainy season (March to May) and short rainy season (October to December). The Tana River transports an estimated 3.1 to 6.8 billion kilograms of sediment annually, significantly influencing the delta's geomorphology and sediment deposition patterns (ASLO Publications). On the other hand, according to bbc.com, Kipini has been experiencing significant coastal erosion, with nearly 10 kilometers of land lost over the past decade due to rising sea levels and sediment dynamics. Developing coastal stabilization strategies for a dynamic coastline like Kipini, which experiences both erosion and accretion, requires a balanced approach that incorporates green-grey solutions.

Figure: ramsar.org

The area that needs to be recognised first for plantation of mangrove is the the deltaic regions where the Tana River meets the Indian Ocean. Then comes the zones between the high and low tide lines along the coastline and previously forested areas now experiencing complete loss.

**Species Selection**: Selecting the appropriate mangrove species for effective coastal stabilization in our study site requires an understanding of the local environmental conditions, including tidal fluctuations, soil types, and salinity levels to choose the species type. Kipini and its surrounding areas host a variety of mangrove species, including *Rhizophora mucronata*, *Avicennia marina*, and *Sonneratia alba*. Salinity Tolerance is measured in Practical Salinity Units (PSU), which quantifies water salinity. *Avicennia marina* is noted for its exceptional tolerance of hypersaline conditions, which can go beyond 45 PSU in some cases.

Figure: From (Left) Avicennia marina, Rhizophora mucronate, Sonneratia alba, Bruguiera gymnorrhiza

| **Species Name** | **Salinity Tolerance** | **Soil Suitability** | **Root Structure** | **Recommended area to plant** |
|---|---|---|---|---|
| *Rhizophora mucronata* | Up to 35 PSU (can tolerate high saline conditions) Alongi, D.M. (2009) | Prefers fine silty to clay soils with good water retention. | Stilt roots extending up to 1.5 m above ground level; aid in wave energy dissipation and sediment trapping | Plant at seaward edge where tidal inundation occurs twice daily to reduce wave impact and stabilize shorelines |
| *Avicennia marina* | 10-45 PSU (extremely salt- | Adapts to sandy loam, saline | Pneumatophores rising 20-30 cm | Ideal for mid-intertidal to upper |

| | | | | |
|---|---|---|---|---|
| | tolerant, adaptable) Clough, B.F (1992) | clay, and mudflats. | above soil for efficient gas exchange, aiding in sediment accretion | intertidal zones where salinity fluctuates with river discharge |
| *Sonneratia alba* | 15-35 PSU (thrives in estuarine salinity gradients) Tomlinson, P.B. (1986) | Sandy loam to muddy substrates, tolerant of fluctuating soil water content. | Buttress roots can reach 1 m in height, providing stability and promoting sediment deposition | Suitable for river-sea interface areas to support sediment stabilization and withstand tidal surges |
| *Bruguiera gymnorrhiza* | 15-30 PSU (moderate salinity tolerance) Ball, M.C (1988) | Best suited for waterlogged, clay-rich, high-organic soils. | Knee roots (plank roots) that help stabilize soil and provide nutrient absorption | Effective in areas with lower wave energy, supplementing other species for layered coastal protection |

By selecting appropriate mangrove species tailored to Kipini's unique conditions, it is possible to achieve effective coastal stabilization, mitigate erosion, and foster habitat restoration.

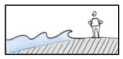
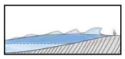
**Planting Techniques**:
Implementing effective planting techniques is crucial for the successful establishment of mangroves in Kipini, ensuring coastal stabilization and ecosystem resilience. The following outlines specific technical methods and considerations for planting mangroves in this unique estuarine environment.

| Planting Technique | Description |
|---|---|
| Direct Seeding | Plant mature propagules directly into soil at a 45-degree angle for stability.<br>Cost-effective and quick.<br>Embed to the right depth to avoid dislodgement.<br>Best For: *Rhizophora mucronata*. |
| Propagation in Nurseries | Grow seeds/propagules for 3–6 months in controlled conditions.<br>Higher initial survival rates.<br>Saline water is used for acclimatization.<br>Best For: *Bruguiera gymnorrhiza*, *Avicennia marina*. |
| Spacing and Density | 1–1.5 meters apart to balance canopy coverage.<br>Cluster Planting: Groups of 5–10 to mimic natural growth.<br>Randomized Planting: Prevents disease spread, simulates nature. |
| Substrate Preparation | Remove debris, keep native vegetation intact to clear the site.<br>Compact soil gently for stability.<br>Add organic matter for sandy areas. |
| Protection and Maintenance | Bamboo or mesh barriers for wave protection.<br>Bi-weekly inspections for growth and pest control.<br>Involve locals for sustainable care. |

**Community Involvement**:
The sites' ecosystem services are of interest to many stakeholders including:

- Local communities who rely on the Delta for food (fish and crops), livestock fooder, water, fuelwood, herbal medicines, timber etc.
- Large scale private agricultural enterprises which will view delta as prime agricultural corner.
- Tana river and Lamu county government.
- The national government agencies ( not limited to Kenya Forest Service (KFS), Kenya Wildlife Service (KWS), National Environmental Management Authority (NEMA) and Water Resources Authority (WRA)
- Conservation NGOs opering in Delta including Nature Kenya, Birdlife International, WWF (World Wide Fund for Nature)
- Community based Organisations (CBOs)

**Analysis of Mangrove Forest against Hazards:**
According to hydrodynamic studies, mangroves reduce wave energy by up to 66% within 100 meters of forest width. Their complex aerial root systems are crucial for dissipating wave energy and slowing water flow, making them effective for erosion prevention and sediment build-up. Even young mangroves can be effective for erosion control and soil stabilization.

| | | HAZARD | | | | |
|---|---|---|---|---|---|---|
| | | Waves  | Storm surges  | Tsunami | Erosion | Sea level rise |
| MANGROVE FOREST PROPERTIES | Width | Hundreds of meters needed to significantly reduce waves (wave height is reduced by 13-66% per 100m of mangroves) | Hundreds of meters needed to significantly reduce wind and waves on top of surge<br><br>Thousands of meters needed to reduce flooding impact (storm surge height is reduced 5 - 50 cm/km) | Hundreds of meters needed to reduce tsunami flood depth by 5 to 30%<br><br>Mangroves do not provide a secure defence (nor do many engineered defences) | Sufficient mangrove forest width needs to be present to maintain sediment balance. This can help to prevent erosion and may encourage active soil build-up. | |
| | Structure | The more obstacles the better: dense aerial root systems and branches help attenuate waves | Open channels and lagoons allow free passage, while dense aerial root systems and canopies obstruct flow | | Complex aerial root systems help slow water flows, allowing sediment to settle and causing sediment to accrete rather than erode. | |
| | Tree Size | Young & small mangroves can already be effective | Smaller trees and shrubs may be overtopped by tsunamis and the very largest storm surges | | Young trees already enable soils to build up. The more biomass input into the soil the better. | |
| | Link to other ecosystems | Sand dunes, barrier islands, saltmarshes, seagrasses and coral reefs can all play an additional role in reducing waves | | | | Allow room for landward retreat of the mangrove |
| | Underpinning factors | Healthy mangroves are a prerequisite for all aspects of coastal protection. Healthy mangroves require: sufficient sediment and fresh water supply and connections with other ecosystems. Conversely, pollution, subsidence (due to deep groundwater/oil extraction or oxidation upon conversion) and unsustainable use jeopardizes mangroves. | | | | |

**Cost-Effectiveness of Mangrove Planting:**
Mangrove restoration is generally cost-effective, offering benefits such as coastal protection, carbon sequestration, and enhanced fisheries. Initial planting cost and maintenance monitorig cost are needed. Initial cost includes collecting seedlings, cultivate them and labor for planting. Maintenance cost includes regular monitoring for some years, fencing protection and training workshop for community. Studies indicate that mangrove restoration can be two to five times

cheaper than engineered structures like breakwaters for wave heights up to half a metre and, within their limits, become more cost effective at greater depths. PLOS Journals

**Potential hazard:**
Relying exclusively on planting without addressing underlying issues may lead to:
Poor Survival Rates: If hydrological conditions are unsuitable, planted mangroves may not thrive. It has to be ensured that natural water flow supports mangroc health. Mangrove Alliance
Invasive Species: Introducing non-native species can disrupt local ecosystems. Ecoshape
Community Resistance: Lack of local involvement can result in neglect or active opposition.
Maintenance and Long-Term Sustainability: Ensuring continued funding and local involvement for monitoring and maintenance is essential. Engaging the community through education and shared benefits can ensure long-term project viability.

**Risk Assessment and Long-Term Projections:**
The integration of natural (green) and artificial (grey) structures creates a multi-layered defense that decreases vulnerability and increases resilience against extreme weather events, such as cyclones and storm surges.

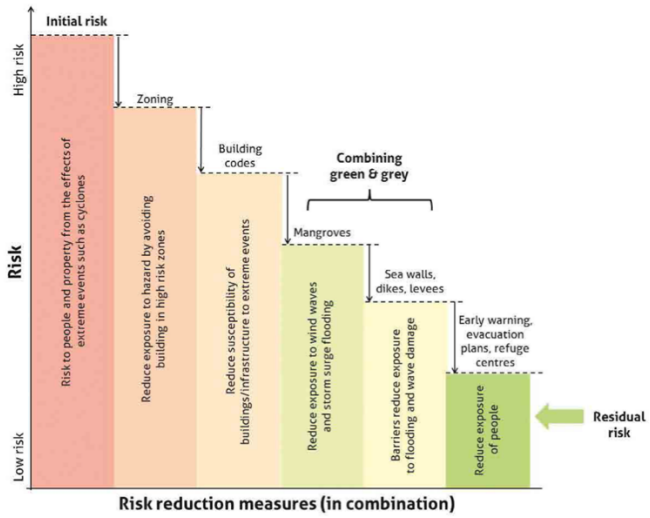


Figure: Risk Reduction measures in combination (US Army Corps of Engineers)

Mangrove forests are known for their long-term stability and resilience, capable of persisting for over 100 years under favorable conditions. The stability of these forests is underpinned by consistent sediment deposition, adequate freshwater supply, and protection from extreme environmental disturbances. However, the long-term survival of mangrove ecosystems depends on effective conservation practices that mitigate human impacts and address challenges posed by climate change. For instance, the Sundarbans mangrove forest has maintained stability over centuries due to continuous sediment input from the Ganges-Brahmaputra delta.(academic.oup.com)
While mangrove forests alone are highly beneficial, combining them with grey infrastructure, such as breakwaters or permeable barriers, can create a hybrid system that provides enhanced protection against extreme weather events.